Linux File System Interview Questions & Answers (DevOps Preparation)

1. WHAT IS A FILE SYSTEM IN LINUX?

A file system is the method Linux uses to store and organize files on a disk. It defines how files are stored, named, and accessed. Common Linux filesystems include ext4, xfs, and btrfs.

Real-time example:

Application logs, databases, and system files on servers are stored using filesystems like ext4.

2. WHAT IS THE ROOT DIRECTORY?

The root directory (/) is the top-level directory in Linux. All other directories originate from it.

Example directories: /home, /etc, /var

Real-time example:

When checking disk usage for the main filesystem we run: df -h /

3. WHAT IS /ETC DIRECTORY?

/etc stores system configuration files.

Examples: /etc/passwd, /etc/fstab, /etc/ssh/sshd_config

Real-time example:

If SSH stops working, engineers check /etc/ssh/sshd_config.

4. WHERE ARE USER HOME DIRECTORIES STORED?

User home directories are stored in /home.

Example: /home/divya

Real-time example:

Developers usually store project files inside their home directory.

5. WHAT IS /VAR DIRECTORY?

/var stores variable data such as logs and application data.

Important location: /var/log

Real-time example:

If an application fails in production, logs are checked in /var/log.

6. WHAT IS /TMP DIRECTORY?

/tmp stores temporary files created by applications. These files may be deleted after reboot.

Real-time example:

Temporary uploads or application cache may be stored here.

7. WHAT IS /DEV IN LINUX?

/dev contains device files representing hardware devices.

Examples: /dev/sda, /dev/sda1, /dev/null

Real-time example:

When a new disk is attached to a server it appears as a device like /dev/xvdb.

8. WHAT IS /PROC?

/proc is a virtual filesystem containing information about running processes and system resources.

Examples: /proc/cpuinfo, /proc/meminfo

Real-time example:

Engineers check CPU information using cat /proc/cpuinfo.

9. WHAT IS THE DIFFERENCE BETWEEN /MEDIA AND /MNT?

/media is used for automatically mounted removable devices such as USB drives.

/mnt is used for temporary manual mounts.

Real-time example:

mount /dev/sdb1 /mnt

10. WHAT IS AN INODE?

An inode stores metadata about a file such as size, owner, permissions, and timestamps. It does not store the filename.

Real-time example:

Sometimes disk space is free but inode usage is full, preventing new files from being created.

11. WHAT IS EXT4 FILESYSTEM?

ext4 is a commonly used Linux journaling filesystem. It supports large files, high performance, and better reliability.

Real-time example:

Many Linux servers use ext4 for application storage.

12. WHAT IS MOUNT IN LINUX?

Mount means attaching a filesystem to a directory so it can be accessed.

Example: mount /dev/sdb1 /mnt

Real-time example:

When a new disk is added to a server it is mounted to store application data.

13. WHAT IS /ETC/FSTAB?

/etc/fstab contains filesystem mount entries that are automatically loaded during system boot.

Example entry: /dev/sdb1 /mnt/data ext4 defaults 0 0

Real-time example:

If an incorrect entry exists the server may fail to boot.

14. DIFFERENCE BETWEEN DF AND DU?

df shows filesystem disk usage.

du shows disk usage of directories and files.

Example commands:

df -h

du -sh /var/log

Real-time example:

du helps find which directory is consuming disk space.

15. WHAT IS FSCK?

fsck checks and repairs filesystem errors.

Example: fsck /dev/sda1

Real-time example:

Used when filesystem corruption occurs after improper shutdown.

SCENARIO 1: DISK IS FULL ON A SERVER

- Step 1: Check disk usage using df -h
- Step 2: Identify large directories using du -sh /*
- Step 3: Check logs inside /var/log
- Step 4: Clear unnecessary logs.

Real-time example:

In production, an application log file was deleted but the application process was still using it. Disk space was released only after restarting the service.

Application logs filled /var/log causing disk full.

SCENARIO 2: DISK SPACE IS FREE BUT NEW FILES CANNOT BE CREATED

- This usually happens when **inode usage is full**.
- Even if disk space is available, if **all inodes are used**, the system cannot create new files.
- Check inode usage - df -i
  Solution: Find directories containing too many small files.
  du --inodes -sh *
  Delete unnecessary files.

Real-time Example:

A monitoring tool created **millions of small log files**, exhausting inodes and preventing file creation.

SCENARIO 3: APPLICATION CANNOT WRITE FILES

- Check dish space using df -h
- Check permissions using ls -l
- Check mount status using mount

Possible causes: disk full, permission issue, filesystem read-only.

SCENARIO 4: DISK ADDED BUT NOT VISIBLE

- Check disks using lsblk
- Check system logs using dmesg
- Verify mount status.

Real-time Example:

When a new disk was attached to an EC2 instance, it appeared as:
/dev/xvdb
We formatted and mounted it to store application data.

SCENARIO 5: SERVER NOT BOOTING AFTER MOUNT

The most common reason is **incorrect configuration in /etc/fstab**.

Linux tries to mount filesystems listed in /etc/fstab during boot.

If the entry is wrong, the system may **fail to boot**.

- Check fstab entry: cat /etc/fstab

- Verify the device exists: lsblk
- Fix the configuration and reboot.

Real-time Example:

A wrong mount entry was added in /etc/fstab, and after reboot the server went into emergency mode.
After fixing the entry, the server booted normally.

SCENARIO 6: FILESYSTEM SUDDENLY BECOMES READ-ONLY

- Check mount status - mount | grep sda
- Check system logs - dmesg
- Repair filesystem - fsck /dev/sda1

Real-time Example:

After a sudden server crash, the filesystem was automatically mounted as **read-only** to prevent further corruption.

SCENARIO 7: **DELETED LOG FILE BUT DISK SPACE IS STILL NOT FREED**

This happens when **a running process still holds the file descriptor**.

Even though the file is deleted, space is not released until the process stops.

- Check open files: lsof | grep deleted
- Fix: Restart the service using the file.

Example:

systemctl restart application

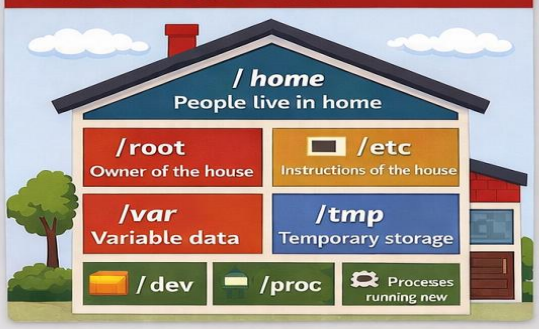
Real-time Example:

An application log file was removed manually, but the process continued writing to the deleted file until the service was restarted.

Easy Tricks to Remember Linux File System

For DevOps Engineers

Trick 1: HOUSE Model



Trick 2: Important Directories

Big Engineers Handle Real **Valuable** Technical DevOps Projects Usually

- **B** /bin → commands
- **E** /etc → config files
- **H** /home → users
- **R** /root → root user
- **V** /var → logs
- **T** /tmp → Temporary files
- **D** /dev → Devices
- **P** /proc → process info
- **U** /usr → software

Trick 3: Static vs Dynamic

STATIC	DYNAMIC
/bin	/var Logs
/usr	/tmp Temp Files

Change daily

Trick 4: Log Location

/var/log VAR = VARIABLE

logs keep changing

/dev/sda

/dev/xvdb

Trick 4: Device Files

/var/log

VAR = VARIABLE = logs keep changing

Trick 5: Device Files

/dev

DEV = DEVICES

- /dev/sda
- /dev/xvdb

Trick 6: Process Info

/proc PROC = PROCESS

/proc/cpuinfo

/proc/meminfo

Trick 8: Just Remember This!

/

- /home – Users
- /etc – Configs
- /var – Logs – Devices
- /dev – Devices
- /tmp – Temp Files